

Alexander Terry

alexterry179@gmail.com • 716-951-0776 • alexanderterry.me • github.com/AlexT01010100 • linkedin.com/in/alexanderterry-computer-science

PROFESSIONAL SUMMARY

Embedded software engineer with experience spanning defense and commercial telecom systems, from low-level C/C++ driver and firmware work through build infrastructure, testing, and CI/CD. Comfortable moving across the stack, from SPI/UART hardware interfaces to containerized backend services and production web applications. Known for finding practical, cost-effective solutions and following projects through from hardware to a working, maintained system.

EDUCATION

The State University of New York at Buffalo

Aug 2021 - May 2025

B.S. Computer Science & B.A. Mathematics | Honors College | GPA: 3.92 / 4.00

Certifications: Google Cybersecurity, IBM Z Xplore Concepts, CITI Biomedical Research Investigators.

WORK EXPERIENCE

Software Engineer | General Dynamics Mission Systems | Scottsdale, AZ

Sep 2025 - Present

- Identified and proposed an open-source alternative to a planned commercial solution, projected to save the program over \$100,000 pending adoption.
- Develop embedded C/C++ software across the full lifecycle, from driver/firmware development through build system tooling, working with cross-compilation toolchains, RTOS/RTEMS environments, and FPGA-integrated hardware within a security controlled defense engineering environment.
- Build and maintain containerized development and build environments, plus Python unit and integration test suites, to standardize embedded builds and support regression testing across build targets.
- Leverage AI coding assistants (ChatGPT, Claude) throughout the development workflow to accelerate implementation, debugging, and code review.

Software Engineer | VOCAL Technologies | Buffalo, NY

Jan 2025 - July 2025

- Developed and optimized embedded C software for real-time voice and data transmission systems (MoIP, VoIP, FoIP), debugging and tuning network protocol stacks (SIP, RTP, TCP/UDP) and audio codecs (G.711, G.729) to meet latency and bandwidth constraints.
- Built custom Yocto Linux images for ARM-based telecom hardware, including initramfs/rootfs boot configuration, U-Boot environment setup, and custom device tree modifications for peripheral support, debugged via UART serial console.
- Hardened deployed systems with iptables firewall rules and AppArmor confinement profiles to minimize attack surface, and automated build and deployment workflows with Perl and Bash to improve CI/CD reliability.
- Implemented and optimized signal modulation and compression techniques at various bitrates to enhance data transmission efficiency and reliability in embedded telecom systems.

PROJECTS

Pi Air Monitor | Multi-sensor telemetry pipeline on a Raspberry Pi 5 | [pi-air-monitor](#)

C, Python, FastAPI, PostgreSQL, Docker, JavaScript

- Wrote C drivers for a BME280 sensor over SPI and an MH-Z19 CO2 sensor over UART, handling calibration, checksum validation, and automatic resync after noisy reads, then exposed both via ctypes to a Python/FastAPI service that samples every 10 seconds, stores readings in PostgreSQL, and serves them through an 8-endpoint REST API, with a JS dashboard showing live/historical data, a 5-tier CO2 danger-level system, and CSV export.
- Added automatic reconnection for both sensors, validation that rejects corrupted CO2 frames and readings above 10,000 ppm, and hourly pruning against a 30-day retention window, then set up GitHub Actions CI to cross-compile for x86_64/aarch64, run the test suite, and publish multi-arch Docker images.

Jaclyn's Threading Website | Full stack booking platform for a local beauty salon | [jaclynbeauty.com](#)

Node.js, Express, PostgreSQL, Docker, JavaScript, HTML/CSS

- Architected and shipped a production booking system end-to-end as sole engineer, handling real-time scheduling, automated notifications, and an authenticated admin dashboard for a live business.
- Hardened the platform with enforced CSP, rate limiting, bcrypt auth, and layered bot mitigation, and eliminated a live double-booking defect via a zero-downtime database migration.

TECHNICAL SKILLS

Programming Languages: C, C++, Python; familiar with Rust.

Embedded Build & Debug Tools: Yocto, Buildroot, U-Boot, Device Tree, Linux Kernel Modules/Drivers, GCC/Clang Cross-Toolchains, Make, CMake, GDB, Valgrind.

Networking & Protocols: SIP, RTP, TCP/IP, UDP, VoIP, FoIP.

Hardware & Interfaces: ARM (Cortex-A/M series), MIPS, UART, I2C, SPI, JTAG, HSDP.

Dev & Lab Tools: Git, GitLab CI, Jira, Docker, Wireshark, Signal Generators, Spectrum Analyzers.